

PROJECT PHASE 1: PROBLEM DEFINITION AND DATA UNDERSTANDING

Deadline: Week 6

Weight: 5% of the project grade

Penalty: Each delay is penalized with **3 points**

Maximum allowed delays: 2

Objective

The goal of this phase is to clearly define the problem, understand the dataset, and position the project within existing research and real-world applications.

Coding (Data Understanding & Exploration)

In this phase, you must load and perform an initial exploration of your dataset.

Minimum requirements:

- Data loading (from source: CSV, API, database, parquet files, etc.)
- Dataset description (number of rows, columns, feature names)
- Data types analysis (numerical, categorical, text, time-series, etc.)
- Missing values analysis
- Basic statistics (mean, median, standard deviation, etc.)
- Distribution analysis (histograms, boxplots, etc.)
- At least **3–5 relevant visualizations** specific to your problem

Report

- Please use the report template provided:
<https://github.com/SergiuLimboi/Intelligent-techniques-for-processing-structured-and-large-data/tree/main/Report%20template>

The report must include:

1. Problem Definition

- Clear description of the problem you aim to solve

2. Importance and Real-World Context

- Why the problem matters
- Where it is used in practice (industry/research)

3. Related Work

- At least **4–5 recent articles** (published in the last 3–4 years)

For each paper, you must include:

- Problem addressed
- Dataset used
- Methodology (models, techniques)
- Evaluation metrics
- Key results

Additionally:

- Provide a **comparison between the approaches**
- Highlight:
 - Advantages
 - Drawbacks

Deliverables (Mandatory)

- **Report:**
 - LaTeX source file
 - PDF version
- **Code:**
 - Jupyter Notebook or Python scripts
 - Must be runnable
- **Repository:**
 - All work must be committed to the **corresponding GitHub Classroom team repository**

For each deliverable upload to GitHub a pdf with the report and a pdf with the code and output (if you use a Jupyter notebook -> export pdf) before the laboratory in which the deadline is.

Grading

- **Report: 50%**
- **Implementation (code & data exploration): 50%**

The final grade for Phase 1 will be the weighted average of the report and implementation. Grades will be provided at the end of the week, after evaluation by the teachers (course and laboratory).

Important

- The work **cannot be evaluated** if it is not committed to the corresponding GitHub team repository.

- Make sure your repository contains **all required files** before the deadline (before laboratory from week 6)

Grading Rubric Code. Data Understanding & Exploration

Coding part. Note: the requirements like “describe”, “interpret”, “comment” etc must be provided as short written text/comments accompanying your code. Extended interpretations will be provided in the report when required

Section	Points	Full points	Partial points	No points
1 . Data loading & setup	1	Data loaded from a valid source in lazy mode using a large data processing tool/library (e.g. polars)	Data loads but not in lazy mode	Data not loaded using a large data processing tool/library
2 . Dataset description	1	Reports rows, columns, feature names, and data shape. Brief contextual description of what the dataset represents.	Shape reported but no feature names or context/what data represents.	No description provided.
3 . Data types analysis	1	Each feature is categorized (numerical, categorical, text, datetime...) with brief justification if features are not categorized by default in the dataset. Type mismatches detected and corrected if the case.	Types listed but no analysis or corrections if needed.	No type analysis.
4 . Missing values analysis	1	Count and find the % of missing values per column reported. Patterns of missingness discussed if the case. Handling missing values strategy mentioned and implemented.	Missing counts, but no % reported compared to all data or discussion of patterns. No or partially handling missing values	Missing values ignored entirely.

Section	Points	Full points	Partial points	No points
5 . Basic statistics	1	Mean, median, std, min, max, and quartiles for numerical features. Frequency counts for categorical columns. Draw insights from the numbers.	Descriptive stats printed but no interpretation.	No statistics computed.
6 . Distribution analysis	2	Histograms and boxplots for key numerical features. Skewness, outliers, and multimodality identified and commented in a few sentences.	Plots present but no interpretation of meaning, shape or outliers.	No distribution plots.
7 . Problem-relevant visualisations	3	At least 3–5 charts directly tied to the research question. Each chart is labelled, titled, and interpreted in 1–2 sentences.	3+ charts present but generic or unlabelled.	Fewer than 3 visualisations, or none relate to the problem.
Bonus	1	Do more than minimum requirements, surprise us with an interesting analysis and insight about the data. You obtained valuable information about data in a creative manner	Tried to do more, but the approach is unclear or not that relevant to the problem. No that valuable insights were obtained	Nothing more than minimum tried
Total	10 points			

Grading Rubric. Report Phase 1

Section / criterion	Points	Full points	Partial points (1/2, 1/3, 1/4 of total points)	No points
1. Problem Definition				
Clear problem description	1	Problem is stated precisely: domain, scope, and objective clearly defined. Reader understands what is being solved without additional context.	Problem described but vague or missing scope/objective.	No clear problem statement, or topic is merely named without explanation.
2. Importance & Real-World Context				
Why the problem matters	1.5	Compelling motivation provided: quantified impact (statistics, scale) or concrete consequences of the unsolved problem.	Motivation mentioned but superficial.	No motivation provided.
Industry / research applications	1.5	At least 2 concrete use cases named (specific industry, company, or research area) with brief explanation of how the solution is applied in practice.	Applications mentioned but too general (e.g. 'used in many fields').	No real-world context given.
3. Related Work				
Number & recency of papers	0.5	4–5+ papers cited, all published within the last 3–4 years, from credible venues (journals, conferences). Proper citations included.	3 papers, or 1–2 papers older than 4 years, or citations missing.	Fewer than 3 papers, or no publication dates verifiable.
Per-paper summary quality	2	For each paper: problem addressed, dataset, methodology, evaluation metrics, and key results all clearly reported. Information is accurate and specific (e.g. exact metric values).	Most components covered but 1–2 items missing per paper, or values are vague ('achieved good accuracy').	Summaries are copy-pasted abstracts, missing most fields, or fewer than 3 papers summarized.
Comparison between approaches	2	Structured comparison across papers on at least 2 dimensions (e.g. dataset size, model type, performance). Comparison is analytical, not just a list.	Comparison present but only describes differences without analysis.	No comparison, papers treated in isolation.
Advantages & drawbacks	1	Each approach's strengths and limitations explicitly discussed. Critique is grounded in evidence	Advantages/drawbacks listed but not discussed.	No advantages or drawbacks listed nor discussed.

Grading Rubric. Report Phase 1				
Section / criterion	Points	Full points	Partial points (1/2, 1/3, ¼ of total points)	No points
Relevance of selected papers		from the paper (e.g. 'limited to small datasets').		
	0.5	All papers directly address the same problem domain, task type, or methodology as the project. The connection between each paper and the project goal is explicitly justified and clear.	Papers are topically related, but some are tangential (e.g. same domain but different tasks), or relevance is assumed without explanation. It is not clear how the selected paper matter for your project	Papers are generic or off-topic (e.g. survey papers unrelated to the specific problem). No justification provided for why the selected papers matter for your project
Tota	10			